

2(a)	any three from: 1 water is a, good / universal / AW, solvent ; A dissolves many substances / description of solvent action A solvent for / dissolves, (named) ions and (named) polar molecules 2 transport of one named substance for a correct reason ; e.g. urea for excretion 3 <u>high specific heat capacity</u> ; A a full description e.g. the amount of heat that must be added to, one unit of mass of the substance / 1 gram or 1 cm³ of water, to increase the temperature by, one unit / 1 °C 4 so temperature of blood remains (fairly) constant / heat is dispersed throughout the body ; <i>must be linked to mp3</i> I temperature of water is constant AVP ; e.g. water is (di)polar	3
2(b)(i)	P - aorta A dorsal aorta Q - vena cava ; I superior and inferior	1
2(b)(ii)	1 aorta / P, transports / delivers / AW, <u>oxygenated blood</u> to the, organs / body / respiring tissues / systemic circulation or aorta / P, transports / AW, blood at high pressure to, organs / body / respiring tissues / systemic circulation ; 2 vena cava / Q, transports / returns / AW, <u>deoxygenated blood</u> to the, right atrium / heart (from systemic circulation) or vena cava / Q, transports / AW, blood at low pressure to the, right atrium / heart ; A vena cava / Q, collects blood from, veins / systemic circulation <i>alternative</i> 3 aorta / P, transports oxygenated blood <u>and</u>, vena cava / Q, transports deoxygenated blood ; 4 blood in aorta to, organs / body / respiring tissues / systemic circulation, <u>and</u> blood in vena cava to, right atrium / heart ;	2

Question	Answer	Marks
2(b)(iii)	<i>closed</i> blood is contained within vessels ; A at least three from heart, arteries / arterioles, veins / venules, capillaries <i>double</i> blood flows twice through the heart in, one / each, (complete), circuit/circulation (of the body) ; A pulmonary circulation and systemic circulation A described	2
2(c)(i)	<i>any two from:</i> (blood at high pressure) will, burst / damage, capillaries ; A cannot withstand high pressure capillaries have thin walls / capillary walls are composed of one (thin) layer of (endothelial) cells ; A capillary walls are one cell thick R capillaries have cell walls <i>idea that</i> at low pressure blood flows slowly to allow, exchange / diffusion, of substances (in capillaries) ; A efficient exchange (of substances) AVP ; e.g. <i>idea that</i> pressure will be low at venous end of capillaries so that tissue fluid is reabsorbed	2

Question	Answer	Marks
2(c)(ii)	any three from: I functions <i>similarities</i> 1 (both have) endothelium / tunica intima ; A tunica interna 2 both have) tunica media / <u>smooth</u> muscle ; <i>differences</i> <i>muscular artery – accept ora for arteriole</i> 3 smaller lumen to wall thickness / ratio of wall thickness to lumen width is smaller ; A thicker wall ; I artery is 'bigger / thicker / wider', unqualified 4 wider lumen ; 5 more <u>smooth</u> muscle (layers) / thick (layer) v thin (layer), <u>smooth</u> muscle ; A thicker tunica media 6 more / presence of, elastic, fibres / tissue ; A no elastic lamina 7 more / presence of, collagen (fibres) ; 8 AVP ; e.g. <i>ref. to</i> blood vessels in wall of artery (vasa vasorum) e.g. no vesicles in endothelium e.g. artery has, tunica externa / adventitia A tunica externa not (clearly) visible in arteriole	3